

NEXUS OSS

Turning any server into a safe, self-cleaning hacking playground

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Quick Terminology: What's a “Challenge”?

One term worth locking in before anything else — everything in this deck builds on it.

Definition

A challenge is a small, deliberately vulnerable program or system — packaged so it can be launched as its own private, running copy for one player at a time.

In CTF (Capture-The-Flag) exercises, hidden inside that program is a “flag” — a secret string the player must find by breaking in. Finding it proves they exploited the vulnerability.

So: one challenge = one type of puzzle. One session = one player's private copy of it.

Example

1

The challenge

“pwn-101” — a small web app with a deliberately broken login page.

2

What the player sees

A link to their own private copy of that web app, and a goal: “find the hidden flag.”

3

What they do

Probe the login page, find the flaw, break in — and uncover the flag hidden inside.

4

What happens after

Their copy is destroyed automatically. The next player gets a fresh, untouched one.

The Problem

CTF (Capture-The-Flag) platforms let hackers practice by attacking planted vulnerabilities. Each player needs their own private, working copy of the challenge.

1

Every player needs their own copy

Not a shared demo — a real, live, breakable environment per person

2

Cloud costs pile up fast

Spinning up cloud VMs for hundreds of players gets expensive quickly

3

Players must not see each other

One player breaking into another's environment defeats the whole exercise

The core question

“How do we give hundreds of people their own isolated playground, without renting hundreds of cloud servers — on hardware we already own?”

That’s exactly what Nexus OSS answers.

Why Not Just Use the Cloud?

Cloud servers (AWS, GCP) look easy at first — but the bill never stops, and it grows with every player.

	vCPU / RAM	Storage	Monthly
AWS t3.medium	2 vCPU / 4 GB	EBS, billed separately	~\$30
GCP e2-medium	2 vCPU (shared) / 4 GB	Persistent disk, billed separately	~\$25
Hetzner bare metal	Dedicated CPU / 64 GB	1 TB NVMe included	~\$59

For roughly double the cost of one small cloud VM, one owned server gives 16x the RAM and includes real storage — and it's a one-time purchase, not rent.

Reference pricing, mid-2026 (Hetzner dedicated, AWS/GCP on-demand list price).

- 1

Cloud VMs that fit the budget don't fit the job

A \$25–30/mo instance has 4 GB RAM — nowhere near enough to run dozens of containers at once.
- 2

Matching the real capacity costs far more

To get 64 GB of RAM on AWS or GCP, the bill jumps to several hundred dollars a month — recurring forever.
- ✓

Bare metal is bought, not rented

One Hetzner-class server at ~\$59/mo (or owned outright) comfortably runs the whole lab — with room to spare.

Start Here

Imagine you're organizing a national hacking **competition**.

500 students. 50 challenges. Every student needs their own isolated environment.

Without Nexus, organizers have to:

- Manually deploy a server for every challenge
- Manually configure networking for every student
- Manually remove instances once they're done
- Manually monitor resource usage the whole time

It's slow. It's expensive. Mistakes happen.

*Nexus is a platform that automatically creates secure practice environments for cybersecurity **competitions** within seconds, manages them while students are using them, and removes them automatically when they're no longer needed.*

Everyone understands that.

Only then explain:

"This is achieved using containers, an isolation layer, and an automated control loop — running on hardware we already own."

What Nexus Actually Does

In plain terms:

Nexus is software that sits on a server and, on request, instantly creates a private, working copy of a challenge, gives one player access to it, and safely destroys it when they're done — all without a human doing any manual setup.

It runs entirely on infrastructure you already own — no cloud subscription required.

Without Nexus

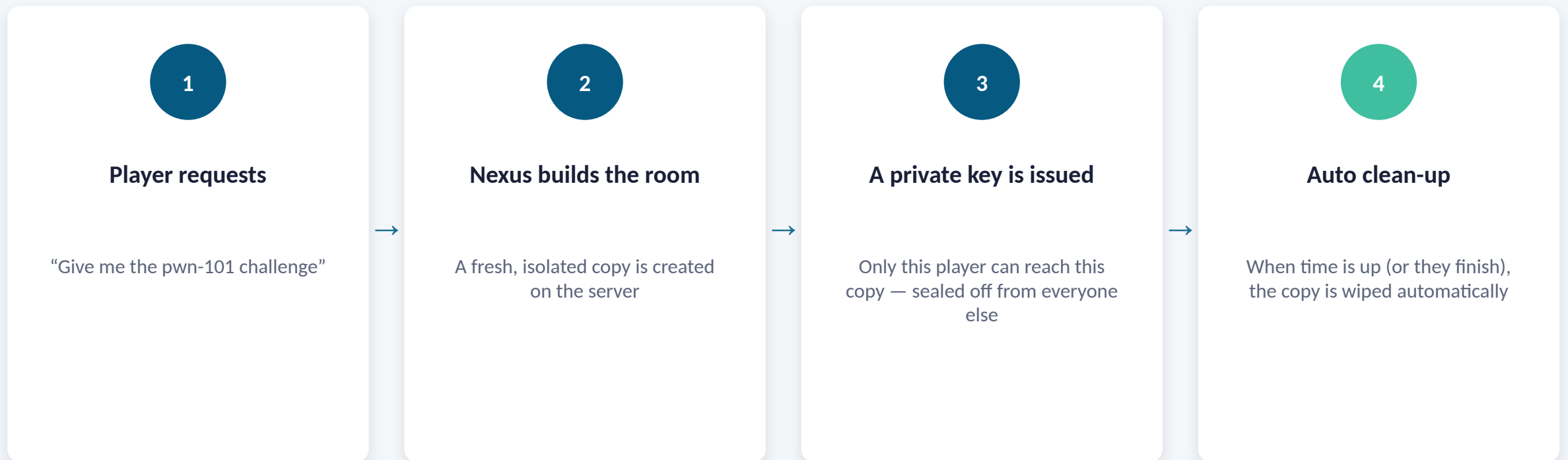
Staff manually set up environments, or the college pays per-hour cloud fees for every session, every player.

With Nexus

One command registers a challenge. From then on, every player session is created, isolated, and cleaned up automatically.

How a Session Happens

From the moment a player clicks “Start Challenge” to having their own private environment — in seconds.



The Four Building Blocks

Nexus is made of four small programs, each with one clear job — like departments in the hotel.

1

The Engine

The brain. Receives requests, decides what to spin up, and keeps track of who has what.

2

The Node Agent

The security guard. Locks each player into their own private lane so no one can reach another's room.

3

The CLI

The control desk. Lets an operator watch everything happening live, in one dashboard.

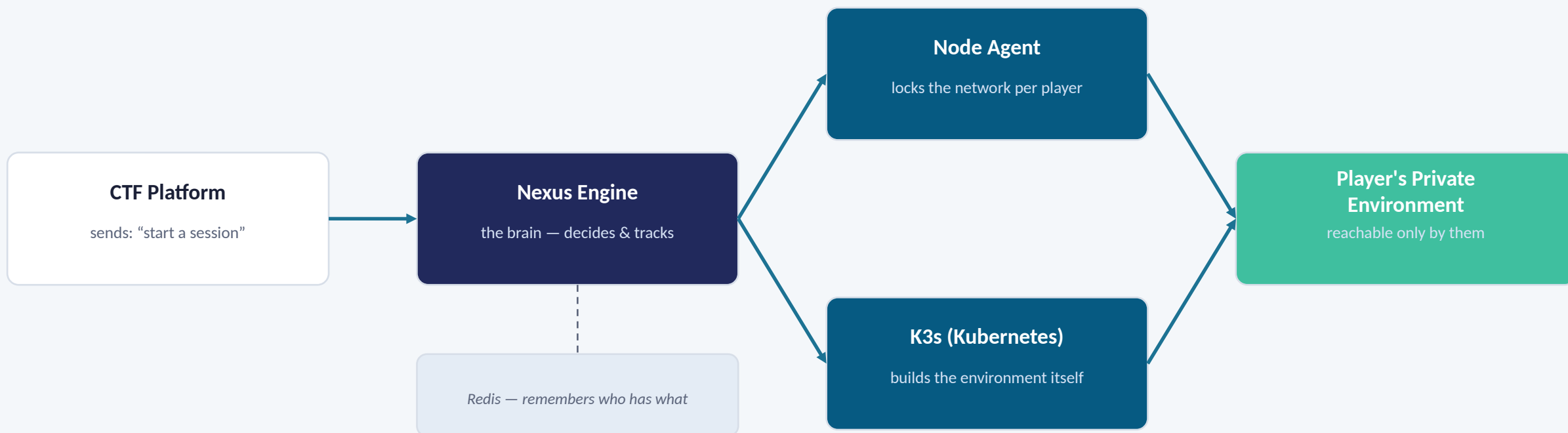
4

The Installer

The setup crew. Prepares a brand-new server for Nexus in one guided run — no manual steps.

How the Pieces Connect

A simplified map of the system — request comes in on the left, a private environment comes out on the right.



Where This Fits, Who Uses It

Nexus was built for CTF platforms, but the same need shows up anywhere people need private, disposable technical environments.

1

College cybersecurity labs

Run student CTF exercises and hands-on courses without a cloud budget

2

Hackathon & CTF organizers

Give every team or player their own isolated environment for the event

3

Corporate security training

Red-team / blue-team drills for employees, run on internal hardware

4

Bug-bounty & CTF bootcamps

Practice ranges for learners, reset automatically after every attempt

Isolation, Not Just Separation

The single hardest problem: making sure no player can ever see or touch another player's environment.

Everyday analogy

Think of a locked mailbox wall. Every resident has a key that opens only their own box. Even standing right next to someone else's box, that key simply will not turn.

That's what Nexus enforces at the network level — automatically, for every single player, every single session.

Private tunnel per player

Each player connects through their own secure, encrypted tunnel into the system.

One-way visibility

A player's tunnel is only ever allowed to reach their own environment — nothing else.

Enforced automatically

These rules are set the instant a session starts and removed the instant it ends — no manual work.

Already Real, Not Just a Prototype

Live at ZecurX

Currently powering a real CTF platform in production

Open source

Published publicly under the Apache 2.0 license

Runs anywhere

Works on any Linux server — laptop, on-prem box, or cloud VM

Why this matters for a college

A department running its own cybersecurity labs, hackathons, or CTF events doesn't need a cloud budget — just one spare server. Nexus turns it into a safe, self-managing practice ground for students.

Questions?

github.com/elish4h · Nexus OSS · Apache 2.0